

SIC AI Chapter Quiz: 04

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Question 01

Option 3: If the significance probability p is greater than the significance level α ($p > \alpha$), the null hypothesis is rejected and the alternative hypothesis is adopted.

Question 02

```
import scipy.stats as stats

# --- Parameters ---
n = 3 # Number of children
p = 0.5 # Probability of a son

# --- 1. Probability of exactly 2 sons ---
# pmf = Probability Mass Function
prob_2_sons = stats.binom.pmf(k=2, n=n, p=p)

# --- 2. Mean ---
mean_val = n * p

# --- 3. Variance ---
variance_val = n * p * (1 - p)

print(f"1) Probability of 2 sons: {prob_2_sons:.3f} (or 3/8)")
print(f"2) Mean: {mean_val}")
print(f"3) Variance: {variance_val}")
```

1) Probability of 2 sons: 0.375 (or 3/8)
2) Mean: 1.5
3) Variance: 0.75

Question 03

Option 2: $(x - \mu)/\sigma$

```
import numpy as np

# Create a sample dataset: Mean=50, Std=10
data = np.array([30, 40, 50, 60, 70])
mu = np.mean(data) # 50.0
sigma = np.std(data) # 14.14...

# Apply Option 2: Standardization Formula
z_scores = (data - mu) / sigma

print(f"Original Data: {data}")
print(f"Mean: {mu}, Std Dev: {sigma:.2f}")
print("-" * 30)
print(f"Standardized (Z-scores): {z_scores}")
print(f"New Mean (should be 0): {np.mean(z_scores):.2f}")
print(f"New Std Dev (should be 1): {np.std(z_scores):.2f}")
```

Original Data: [30 40 50 60 70]
Mean: 50.0, Std Dev: 14.14

Standardized (Z-scores): [-1.41421356 -0.70710678 0. 0.70710678 1.41421356]
New Mean (should be 0): 0.00
New Std Dev (should be 1): 1.00

Question 04

Option 2: χ^2

Question 05

Answer: The 95% confidence interval is approximately $78.93 \leq \mu \leq 81.07$.

```

import numpy as np
import scipy.stats as stats

# --- Given Data ---
n = 30
mean = 80
variance = 9
std_dev = np.sqrt(variance) # Square root of 9 is 3

# --- Method 1: Z-Distribution (Standard for n >= 30) ---
z_score = 1.96 # Standard critical value for 95%
margin_of_error_z = z_score * (std_dev / np.sqrt(n))

lower_z = mean - margin_of_error_z
upper_z = mean + margin_of_error_z

print(f"95% Confidence Interval (Z-score 1.96):")
print(f"[{lower_z:.2f}, {upper_z:.2f}]")
print("-" * 30)

# --- Method 2: T-Distribution (More precise for samples) ---
# Degrees of freedom = n - 1 = 29
t_score = stats.t.ppf(0.975, df=n-1)
margin_of_error_t = t_score * (std_dev / np.sqrt(n))

lower_t = mean - margin_of_error_t
upper_t = mean + margin_of_error_t

print(f"95% Confidence Interval (T-score {t_score:.3f}):")
print(f"[{lower_t:.2f}, {upper_t:.2f}]")

```

```

95% Confidence Interval (Z-score 1.96):
[78.93, 81.07]
-----
95% Confidence Interval (T-score 2.045):
[78.88, 81.12]

```

Question 06

```

import numpy as np
import pandas as pd
from sklearn.datasets import load_breast_cancer

# 1. Load data
data = load_breast_cancer()

# 2. Make DataFrame (Correcting syntax for columns)
df = pd.DataFrame(data.data, columns=data.feature_names)

# 3. Assign Target (Inverted as per the question)
# Original: 0=Malignant, 1=Benign
# New: 1=Malignant, 0=Benign
df['target'] = 1 - data.target

# 4. Calculate Correlation with 'target'
# We drop the 'target' column itself so it doesn't compare with itself (which is always 1.0)
correlations = df.corr()['target'].drop('target')

# 5. Find the feature with the highest correlation
most_correlated_feature = correlations.idxmax()
max_value = correlations.max()

print("--- Correlation Results (Top 5) ---")
print(correlations.sort_values(ascending=False).head())
print("-" * 40)
print(f"The feature most correlated with the target is: '{most_correlated_feature}'")
print(f"Correlation Coefficient: {max_value:.4f}")

```

```

--- Correlation Results (Top 5) ---
worst concave points    0.793566
worst perimeter         0.782914
mean concave points     0.776614
worst radius            0.776454
mean perimeter          0.742636
Name: target, dtype: float64
-----
The feature most correlated with the target is: 'worst concave points'
Correlation Coefficient: 0.7936

```